

6. Template Attack

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The aim of the exercise is to perform a profiled attack (Template Attack) targeting the Hamming weight of the output of the first round of the AES S-box. The goal is to reveal the first byte of the encryption key.

1. Download the prepared data

[Here you can find a ZIP archive with the data.](https://oleksak.net/fit/ni-hsc-lab-06-data.zip) (<https://oleksak.net/fit/ni-hsc-lab-06-data.zip>) In Google Chrome you may need to open the link in a new tab.

The archive contains two sets of data: a profiling/training set, and a set for the attack itself.

- The profiling_keys.uint8.bin file contains 6,000 encryption keys (6,000 x 16 bytes)
- The profiling_plaintext.uint8.bin file contains 6,000 plaintexts (6,000 x 16 bytes)
- The profiling_traces.double.bin file contains 6,000 consumption traces, each consisting of 5,000 samples in **double** format, i.e. floating point with double precision (6,000 x 5,000 x 8 bytes)
- The file attack_plaintext.uint8.bin contains 20 plaintexts (20 x 16 bytes)
- The file attack_traces.double.bin contains 20 consumption traces, each consisting of 5.000 samples in **double** format, i.e. floating point with double precision (20 x 5.000 x 8 bytes)
- The data are stored in "binary format", little-endian. This is the format you expect if you are working on a classic PC (Intel, AMD).

2. Create the templates

The goal is to create templates that will characterize Hamming weight of the first S-box output in the first round. To do this, use the profiling set. Work only with the first byte!

2.1. Partition the consumption traces

Compute $f(x_i) = \text{Sbox}(\text{plaintext XOR key})$ for every trace, and partition the traces into nine groups (HW=0, ..., HW=8) according to the Hamming weight of $f(x_i)$.

2.2. Select the points of the interest

For each of the nine groups, calculate the average consumption trace. Then, using the Sum of Differences method, select 5 (in words, five) suitable sampling points. Continue to work with these points only.

2.3. Compute the Gaussian parameters

For each of the nine groups, calculate the average for each point (already done from step 2.2). Subtract these average traces from the individual traces to get only the "noise" and calculate its covariance matrices (5x5).

3. Use the prepared templates for the attack

Use a set with 20 traces and plaintexts, dedicated for the attack. Again, limit yourself to the previously selected sampling points (points of the interest). You will probably need significantly fewer than 20 traces to successfully reveal the key.

1. For each of the 256 key candidates, and each plaintext, compute the hypothetical Hamming weight of the output of the first S-box.
2. Using the templates created earlier, determine the probability that the corresponding trace was measured with the given Hamming weight
3. Calculate the logarithm of that probability, and add it to the overall probability for that key candidate (if we weren't using the logarithm, we would multiply instead of adding)
4. As a result, choose the most likely key candidate

4. Submit the results in the form of a brief report

If you work in Jupyter, Mathematica, Matlab, you can export your notebook to PDF. If you don't have any such export, screenshots will suffice.

The chosen workflow must be clear from the submitted report. The report must contain all the required graphs (with appropriate comments) and key values.

Submit the report through the submitting system EET ([CZ \(https://ddd.fit.cvut.cz/EET/hsc/index.php\)](https://ddd.fit.cvut.cz/EET/hsc/index.php)) ([EN \(https://ddd.fit.cvut.cz/EET/en-hsc/index.php\)](https://ddd.fit.cvut.cz/EET/en-hsc/index.php)) no later than the next but one lab (November 18, 2025).

If you have any additional questions or consultations, or you simply get stuck, please do not hesitate to contact us by email or MS Teams.